Edge-IoT and MLLMs for Education and Scene Understanding: Assisting Vision and Hearing-Impaired Individuals

Mourad Raif, Hasna Chaibi, Nordine Quadar, Abdessamad El Rharras, and Rachid Saadane

Abstract

The rapid advancement of IoT and edge computing technologies has opened new horizons for creating assistive solutions tailored to individuals with sensory impairments, particularly those with hearing and vision disabilities. This article presents a novel edge-IoT-based framework that integrates multimodal large language models (MLLMs), multi-object tracking, and scene understanding to develop real-time, responsive assistance for impaired individuals. Our proposed system is designed to enhance accessibility and quality of life by providing educational tools and entertainment options that cater specifically to the needs of this community. The proposed solution leverages the computational power of edge devices to process data locally, ensuring low latency and high responsiveness, which are critical for real-time applications. Furthermore, we explore the potential of generative AI models in improving autonomy, with a particular focus on real-time transcription services for the hearing impaired and scene description services for the visually impaired. This work demonstrates the feasibility and effectiveness of using edge-IoT technologies combined with advanced AI techniques to create inclusive environments that empower disabled individuals, ensuring that technological advancements are leveraged to foster accessibility and independence.

Introduction

The integration of Artificial Intelligence (AI) and the Internet of Things (IoT) is transforming the technological landscape, creating unprecedented opportunities for innovation. AI's ability to process vast amounts of data and generate insights, coupled with IoT's capability to collect real-time data from a multitude of sources, is revolutionizing various industries.

This convergence is facilitating smarter environments that can anticipate and respond to needs with remarkable precision. However, despite these advancements, the value of human-based skills cannot be overstated. While AI and IoT systems excel in data processing and operational efficiency, human intuition, creativity, and empathy remain irreplaceable. The nuanced understanding and emotional intelligence that humans bring to problem-solving and decision-making are critical for areas where technology alone falls short.

For example, in education, Generative-AI is a tool to foster deep connection between educators and learners who suffer from hearing impairment. Also, scene understanding AI-tools, support autonomy, emotional and cognitive growth of the people with visual impairment, towards their surroundings.

The key to maximizing the benefits of AI and IoT lies in harmonizing these technologies with human-based skills. This balanced approach ensures that while we leverage AI for efficiency and IoT for connectivity, human judgment and adaptability remain at the core of innovation. The future of technological progress depends on this synergy, fostering environments where AI and IoT enhance, rather than replace human expertise.

Based on this fact, holistic and human-based frameworks are crucial to align research development with a good public understanding of opportunities, empower people, and pursue equity access in general, especially for people with disabilities.

Most industry solutions are intended for people without disabilities; our article on access control and intrusion detection is an example [1], the motivation behind our study is to deliver two use-cases of Generative AI usage specifically tailored for this population.

The main contributions of this study can be summarized into two aspects:

Technical: Over the years, research efforts have focused on adapting IoT physical systems, using the power of blockchain for optimal security in solution design. However, to our knowledge, a formal edge-IoT framework compliant with Industry 5.0 has yet to be designed in academic research. Through a use case, this article explores the potential of edge IoT-based technologies to efficiently implement the principles and components of Industry 5.0 at a low cost. We will briefly introduce the latest cutting-edge techniques in the field of generative AI and discover the recent ways it is used in multi-object tracking tasks and

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scenes understanding, audio, and text processing. All these techniques and algorithms will be blended into one solution ready to scale up.

Social awareness: Anticipating the future, the primary challenge for Edge-IoT-based products and services will be the development of practical solutions, fostering collaboration, and ensuring user-friendly assistance. This is particularly crucial for people with disabilities. According to the World Health Organization (WHO), by 2050, there will be 700 million individuals with hearing disabilities, with 80% living in low-income countries [2].

Furthermore, 25% of people over 60 years of age have hearing loss. According to Georgetown University, 18% of the 65+ US population suffer from visual impairment that cannot be corrected with ordinary glasses, contact lenses, medical or surgery [3]. There are many solutions based on phone speech, sign language, or tactile signing, designed to help individuals with impaired vision and hearing to have a maximum of autonomy. Through this work, we seek to raise awareness among the public that new technologies are opening new horizons for this community, providing a more active experience, and providing affordable technologies for retention or scene understanding.

Most solutions focus on digital accessibility; as can be witnessed in the Harvard accessibility policy, retention is not part of the process. We provide a solution and an opportunity for educational institutions, assistive services, and cinema production companies to take their products and services to the next level.

The structure of this article is as follows: We delve into the literature of Edge-IoT applications for individuals with disabilities, particularly those with vision and hearing impairments. Also, this section examines the recent developments regarding the use of Multi-Modal LLMs in education and entertainment. We introduce our use case, presents the results, and discusses the developed solution. We cover the discussion and limitations. We conclude the article and outlines future work.

Related Work

Multimodal LLMs in Education and Entertainment

The integration of Large Multimodal Models (LMMs) in education and entertainment represents a transformative shift in accessibility, particularly for individuals with disabilities. Recent models, such as Kosmos-2 and Flamingo have demonstrated significant improvements in video captions, for people with hearing impairment, with high performance of ChatGPT regarding baseline evaluation systems. For instance, Fathallah *et al.* [4] achieve a BLEU (Bilingual Evaluation Understudy, to calculate the number of correct predictions of the model, compared to a ground truth captions) score of 67% on YouTube-ASR, 68% on Llama2-13B, and 85% on ChatGPT, opening the field for more research improvements. The benefits of LLMs extend beyond hearing accessibility, significantly enhancing scene understanding and real-time processing for visually impaired individuals. Kondandaram *et al.* [5] demonstrate how LLMs improve interaction with complex digital interfaces for blind users. Their system, Savant,

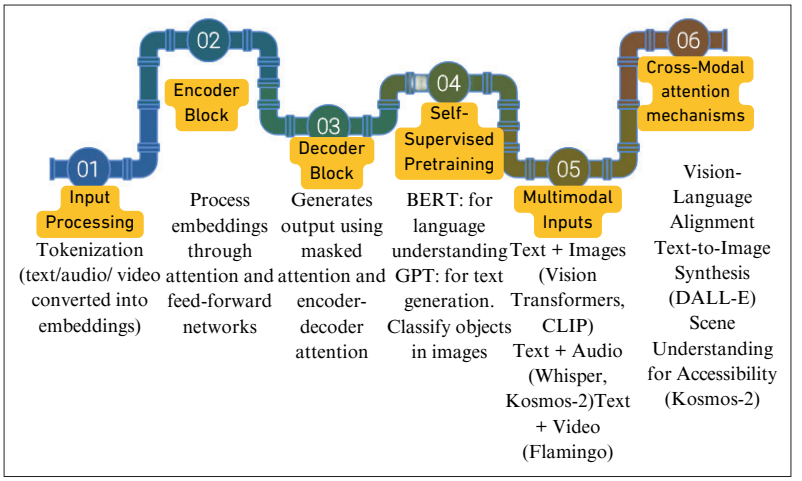


FIGURE 1. From transformer models to LLMs and MLLMs.

enables users to control graphical user interfaces (GUIs) via natural language commands, eliminating the reliance on cumbersome screen reader navigation. Their study leverages the power of LLMs, by increasing x4 times the task completion speed. Also, LLM-assisted vision accessibility achieves a BLEU score of 89% compared to 64% with OCR-based methods, while improving object detection precision from 72% to 91%.

Figure 1 represents a walkthrough of the entire process, involving the usage of transformer-based algorithms, toward conversational results, regarding text, speech and image processing.

We can clearly see the transformative power of transformer-based architectures in Generative AI. It enables scalable learning, efficient text generation, and advanced vision-language alignment for real-world applications.

The application of MLLM in education, especially for people with disabilities, is still an emerging field. Bewersdorff *et al.* [6] explore the potential of MLLMs to revolutionize education by providing personalized learning experiences. The study emphasizes the importance of integrating MLLMs into educational systems without overshadowing the role of educators, suggesting a balanced approach where MLLMs serve as assistive tools rather than replacements for human teachers. This is particularly relevant for students with disabilities, as these models can adapt content to various needs, offering alternative explanations or formats that are more accessible.

Wang *et al.* [7] provide a detailed examination of the reasoning capabilities of MLLM, highlighting their potential to improve learning experiences for disabled people. The study discusses how instruction tuning, prompt design, and using external tools such as Python can enhance MLLMs' reasoning abilities. For visually impaired individuals, these models can offer detailed descriptions and explanations of visual content, making educational materials more accessible.

In the context of entertainment, Argaw *et al.* [8] and Liu *et al.* [9] address the challenge of scene understanding in videos, which is crucial for creating accessible content for visually impaired audiences. They focus on leveraging MLLMs to process and interpret long-form video content, providing a more immersive experience for visually impaired viewers. Their work builds on the

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temporal segments-of-interest approach, allowing for a nuanced understanding of complex visual narratives in movies and TV shows.

Fan *et al.* [10] introduce VideoAgent, a multimodal agent designed to improve video understanding without extensive training data. This tool is particularly useful for visually impaired people as it can generate detailed and accessible summaries of video content. By applying such technologies, entertainment providers can offer a more inclusive experience for visually impaired viewers.

In another study, Yang *et al.* [11] present VIAssist, a system specifically designed for visually impaired users. VIAssist adapts MLLMs to the unique needs of visually impaired individuals by providing reliable visual-question answering (VQA) capabilities. The system guides users through capturing better-quality images and accurately answers their queries. This approach significantly improves the usability of MLLMs for visually impaired users, allowing them to interact more effectively with visual content in both educational and entertainment settings.

Advancements in IoT and Edge Computing for Disabled Individuals

The incorporation of IoT and edge computing technologies into assistive systems for disabled individuals represents a major leap forward in enhancing their autonomy, and hence their quality of life. By processing data locally at the edge, these systems significantly reduce latency, bolster security, and offer real-time responsiveness, an essential component in assistive technologies.

Real-Time Processing and Secure Communication: Edge computing enables real-time processing, which is a cornerstone of modern assistive technologies. For example, in communication assistance for people with severe disabilities, such as those who are paralyzed, the ability to interpret and transmit neural signals quickly and accurately is paramount.

Rajesh *et al.* [12] demonstrated that edge computing can facilitate secure brain-brain communication by processing the electroencephalography signals (EEG) at the edge, thus ensuring that sensitive data are transmitted securely and promptly between the patient and the caregiver. This approach enhances the efficiency of these assistive systems and ensures that critical communication occurs without delays, which could compromise the user's safety and well-being.

Security is another crucial advantage provided by edge computing. By handling data encryption and decryption locally, edge computing minimizes the risks of transmitting sensitive information over potentially insecure networks. This is particularly important for assistive devices, as these devices often handle personal and sensitive data. Rajesh *et al.* [12] highlighted how their proposed Novel Tiny Symmetric Algorithm (NTSA) efficiently encrypts data at the edge, ensuring secure communication in IoT-enabled assistive systems.

Accessibility and Quality of Life Enhancement: Edge computing also plays a pivotal role in improving accessibility and ensuring a higher quality of life for disabled individuals. Rathee *et al.* [13] emphasized the importance of designing edge computing systems that prioritize accessibility and are designed to meet the unique needs of disabled users. Their research focused on creating trusted edge computing frameworks that not only enhance the functionality of assistive devices but also ensure that these systems are user friendly and accessible to all, regardless of the user's specific impairments.

By reducing dependency on centralized cloud services, edge computing allows for more personalized and adaptive responses from assistive technologies. For example, wearable devices that assist visually impaired people can take advantage of edge computing to process real-time video data, providing immediate and contextually relevant feedback. This enhances the user's situational awareness and autonomy, allowing them to navigate their environment more effectively.

Energy Efficiency and Portability: Energy efficiency is another critical factor in deploying IoT and edge computing in assistive technologies. These devices, often portable and used in various environments, require efficient energy management to function effectively over extended periods. By processing data locally, Edge computing frameworks reduce the need for constant communication with cloud servers, thereby conserving energy and extending battery life.

Rathee *et al.* [13] also discussed how edge computing can be optimized for low-power devices, making these technologies more viable for daily use by disabled individuals. This focus on energy efficiency enhances the usability of these devices and ensures that they remain operational in situations where access to power sources is limited.

Advances in IoT and edge computing are transforming assistive technologies for disabled individuals, making them more secure, responsive, and energy efficient. By enabling real-time processing, improving data security, and improving energy management, these technologies significantly improve the quality of life for disabled users. As research in this field progresses, it will be crucial to optimize these systems further to meet the specific needs of disabled individuals, ensuring that the full potential of IoT and edge computing is realized in creating more inclusive and accessible environments.

Use Case: Edge-Enabled Generative AI for Disabled People Assistance

Overview of the Solution

In our solution, Fig. 2a and Fig. 2b we will address hearing and vision impairment due to aging or health deterioration, ensuring that people's brains have all vision and hearing abilities and that our solution will assist them in maintaining them. This has the advantage of keeping elderly people and people with deteriorated vision and hearing sharp and in good control of their lives. We suppose that a person suffers from only one disability, hearing or vision.

For impaired hearing individuals, we produce a transcription and multiple-choice generation to assist them in getting a good understanding of educational content. For individuals with impaired vision, we provide a framework that entertainment production companies can use to offer a new cinema experience to this population.

Methodologies, Architecture, and Material

Methodology:

Data:

- For impaired hearing individuals, we simulated an inspirational speech made by Chris Anderson, the founder of TED Talks, about the key points for delivering high-impact public presentations. For evaluations, we used the CommonVoice-fr dataset.
- For impaired vision individuals, we took a sequence from the movie "Casablanca" for the proof of concept.

Theoretical and practical tools:

- *Large language models (LLMs):* these are frameworks based on training of large deep-learning neural networks. Their main contribution is their ability to substantially promote conversational systems, based on the enormous volume of training data used.
- *Multimodal Large Language Models (MLLMs):* These are models which scaled up LLMs from Natural Language Processing (NLP) to image and video processing. In our solution, we used:
  - *Whisper:* a large language model from openAI responsible for speech transcription into text. We use its distilled versions: tiny, small and base;
  - *Mixtral:* it is the cornerstone LLM from the French AI company Mistral, responsible for compiling text and reading Quizzes for retention and understanding purposes.
- Transformers: They represent the backbone of large language models based on the notion of tokenization, for models long term memory retention.
- *Haggingface:* this is a collaborative platform, filled with available models to use, ranging from speech to text, text to speech, image to image and any use case possible through large models.
- *Object detection and multi-object-tracking techniques:* these are techniques responsible for detecting and fallowing objects through frames, for scenes understanding.
- *Moviepy:* A Python library for automated video edition.

Solution Architecture: Figure 2a and Fig. 2b represent the pipelines, incorporating individuals with hearing and vision impairments in our solution.

For a population with vision disabilities, the biggest challenge is to apprehend the surrounding context and video content. Our solution has many applications, including interactive scene understanding and movie description. For example, in public spaces, people can choose when to record a video and pass it to the pipeline for processing. The audio will be directed to one ear. The video flow description will be directed to the other ear after applying a specialized LLM such as Kosmos2 and then transcribed using another text-to-speech LLM, the FastSpeech2 model. We will provide a more specific schema for movie processing Fig. 4 and Fig. 5.

For impaired Hearing, let's imagine a classroom with students with hearing impairment. While the teacher presents his content, many points might raise comprehension concerns for them, resulting in discomfort and confusion, leading to school drawbacks.

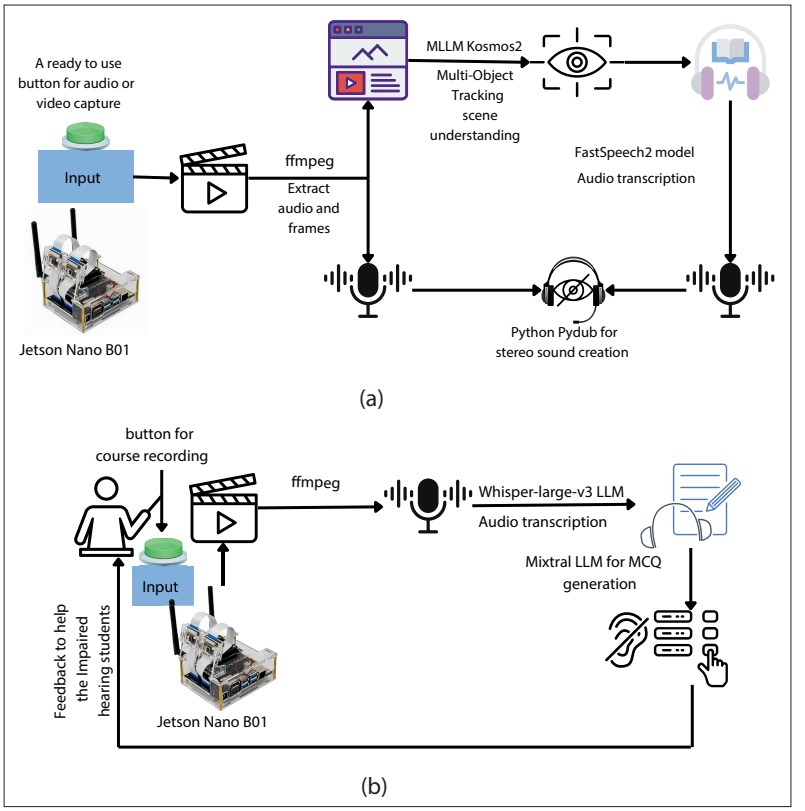


FIGURE 2. a) Conceptual solution design for impaired vision persons; b) conceptual solution design for impaired hearing persons.

	Model	Parameter count	BLEU	WER
Our solution	Tiny	~39 million	42.13	0.42
	Base	~74 million	57.12	0.29
	Small	~244 million	73.17	0.17
[4]	Llama2	13B	67.00	0.24
[4]	ChatGPT	175B	85.00	0.098

TABLE 1. Solution results vs. Llama2 and ChatGPT.

Device	Tiny	Base	Small
AWS instance (A10G)	~180 Wh	~200 Wh	~250 Wh
Jetson nano B01	~10Wh	~12 Wh	~15 Wh

TABLE 2. Energy consumption on the edge and cloud.

Our solution provides the teacher with a button to choose the important sections that his microphone will record and send to whisper model for transcription. Then, a multiple-choice questionnaire will be generated with Mixtral LLM to test student retention. The teacher has a dashboard to check students' comprehension for further help Fig. 2b.

Our solution and previous analysis provide a new opportunity for this population, which can be scaled up. With further development we can provide a prototype with acceptable dimensions. Considering the amount of computing large language models need, a GPU-based architecture is necessary. We opt for an Amazon Web Service machine with 1 Nvidia A10G GPU and 8CPUs. The jetson nano B01 role is to offload heavy processing to the AWS instance.



FIGURE 3. The result for speech understanding assistance

Results for Impaired Hearing solution

Figure 3 presents the output of our final solution, following its application to a video input for multiple-choice understanding tests. We selected the Flask framework for the end user due to its ease of implementation. For future iterations, Django or another advanced framework would provide a more robust interface for development.

Our small ASR model achieves state-of-the-art BLEU and WER compared to Llama2, while being much smaller and more efficient. Against, ChatGPT our model provides 86% of its BLEU performance with less than 1% of its parameter count, providing the efficiency of domain-specific ASR models.

Also, the energy efficiency of our Edge deployment is 20 times more energy-efficient than AWS.

Results for Impaired Vision solution

In this section, we delve into the intricacies of scenes under supervised conditions during movie production. Further steps are required to achieve automation Figure 4, including integrating post-production cut-scene files, which can be processed using the Python library MoviePy and then fed into the processing pipeline shown in Figure 2a. The end-user solution will feature stereo output for individuals with hearing impairments. The strength of this solution lies in its ability to process large volumes of archival information at a lower cost.

However, the system described above, using Kosmos2 for scene description, is only partially self-sufficient, as illustrated in Figure 5.

The MLLM responds to the question: "What is/are the actor(s)/actress(es) doing in this picture?."

To enhance this, our approach involves using supervised learning for object detection and tracking before submitting the results to Kosmos2. We can employ lightweight face detection algorithms like HOG or Eigenface, but challenges will persist for scenes with adverse weather conditions, such as rain or snow, as highlighted in [1], which reported a 30% drop of accuracy under severe weather conditions.

Figure 5 shows that despite the innovative focus of new MLLMs on image processing, like Kosmos2, many challenges can still arise. The combination of these new techniques with well-established algorithms, like face recognition and multi-object tracking, can have a tremendous impact on future works.

We intend to investigate in depth the parameters which result in some actors' automatic recognition and others only with face recognition.

Discussion and Future Work

The integration of IoT and edge computing has shown tremendous potential in transforming assistive technologies for people with disabilities, particularly those with hearing and vision impairments. Our research demonstrates that leveraging an edge-IoT-based framework incorporating MLLMs, multi-object tracking, and scene understanding can significantly enhance accessibility and quality of life for this population. Processing data locally at the edge ensures low latency and high responsiveness, which are crucial for real-time assistance and empowering user autonomy.

However, the current system is not without challenges. A notable limitation we encountered was the processing bottleneck when processing whisper-medium or large. But the good news is the competitive transcription accuracy with BLEU scores of 42.13 (Tiny), 57.12 (Base), and 73.17 (small) and corresponding WER values of, 0.42, 0.29, and 0.17 respectively. Compared to large state-of-the-art models like Llama2 (67 Bleu, 0.24 WER) and ChatGPT (85 BLEU, 0.098 WER), our lightweight models achieve competitive accuracy while requiring significantly fewer computational resources.

The bottleneck computation highlights the need for further optimization, in architecture, resources or algorithmic choices, as real-time applications demand more efficient processing. Addressing this issue requires a combination of more powerful hardware, refined algorithms, and software optimized for edge computing environments.

Energy efficiency remains a challenge, particularly given the computational demands of AI models integrated into portable devices. Although edge computing reduces reliance on

cloud communication, the energy required to run these models can be substantial, 250Wh on AWS against 15Wh on Jetson Nano for the Whisper small model, for example. Future efforts should focus on refining these models to operate effectively within the constraints of edge devices, by developing lightweight versions that maintain functionality while consuming less energy.

Despite these hurdles, the potential benefits of this technology are substantial. Real-time transcription for impaired hearing people and scene descriptions for the visually impaired people offer significant advances in accessibility, particularly in educational and entertainment contexts. In addition, the framework we developed could be extended to support other types of impairments, providing new tools for navigation, learning, and more.

Future research should focus on addressing these limitations by developing more efficient algorithms, enhancing hardware capabilities, and further optimizing the system for real-time applications. It will also be crucial to expand use cases to cover a broader spectrum of disabilities and to adapt technology to diverse environments, particularly those with limited resources, to ensure widespread adoption.

Conclusion

In conclusion, while our system demonstrates the transformative potential of IoT and edge computing in the realm of assistive technologies, it also underscores the need for continuous optimization in energy efficiency, computational cost, and real-time processing. Our approach achieves state-of-the-art transcription accuracy with significantly reduced computational demands, proving that smaller, domain-specific AI models can rival large-scale models while operating efficiently on low-power devices. By effectively addressing and overcoming the current technical challenges, we can propel these technologies forward, fostering the creation of more inclusive and empowering solutions specifically tailored for impaired individuals. This advancement will significantly contribute to enhance their independence and overall quality of life, making technology an even more powerful tool for societal improvement.

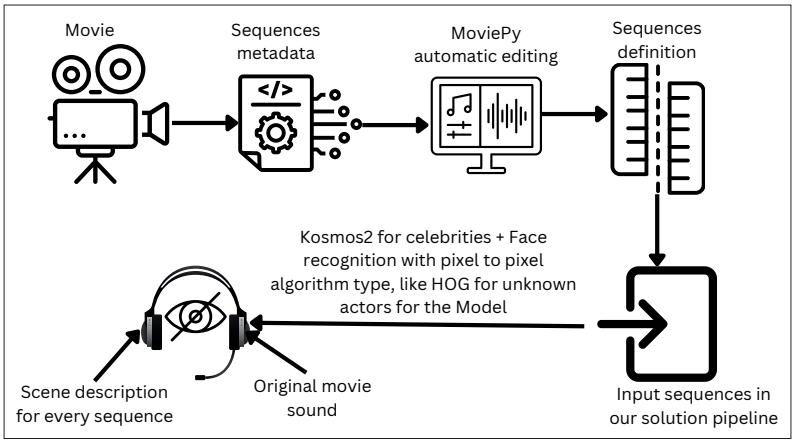


FIGURE 4. Automatic movie scenes understanding.

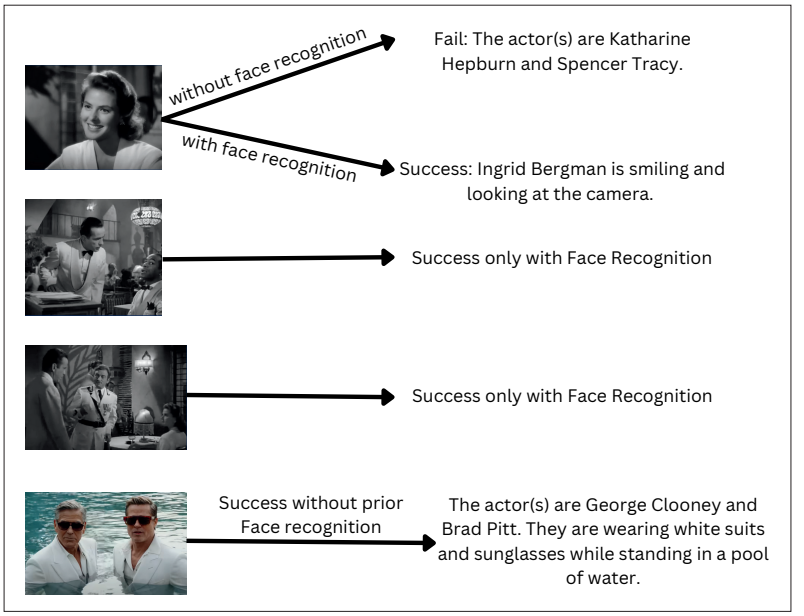


FIGURE 5. Kosmos2 video processing: with vs. without face detection

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